DO NOW — RECALL

Run-length and dictionary coding

Name: _____ Class: _____ Date: _____

1. Define lossless compression.

2. What is a run in RLE?

3. What is a compression overhead?

4. For array [2,3,4,5], double values at indexes 1 and 3. State final array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Original can be reconstructed exactly.

2. Model answer

Consecutive repeated instances of the same symbol.

3. Model answer

Extra metadata needed to represent/decode data.

4. Model answer

[2,6,4,10], sum 22; indexes are positions 1 and 3.

Vocabulary — know these terms

RLE — Encode repeated runs with counts.

run — Maximal consecutive equal symbols.

dictionary coding — Represent repeated strings by references.

dictionary — Mapping of codes to values/phrases.

overhead — Metadata/structure required by encoding.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Decoder reads both references with its matching dictionary and reconstructs the original phrase.

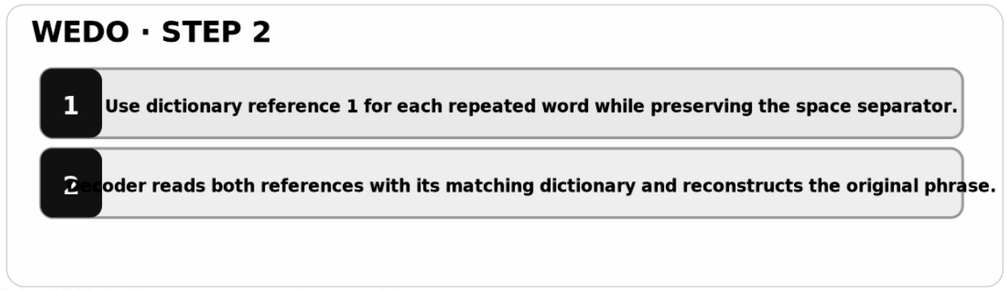
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A dictionary maps index 1 to 'RED'. Encode 'RED RED' and decode the result.

Use dictionary reference 1 for each repeated word while preserving the space separator.



WEDO · STEP 2: Use dictionary reference 1 for each repeated word while preserving the space separator.; Decoder reads both references with its matching dictionary and reconstructs the original phrase.

NEW LEARNING

Read, annotate and explain

Understand

Run-length encoding replaces each maximal run of the same symbol with a value and count; it works best for long consecutive repetitions.

Apply

Dictionary coding replaces repeated strings with dictionary references; the decoder needs the same dictionary, transmitted or reconstructed according to the scheme.

Evaluate

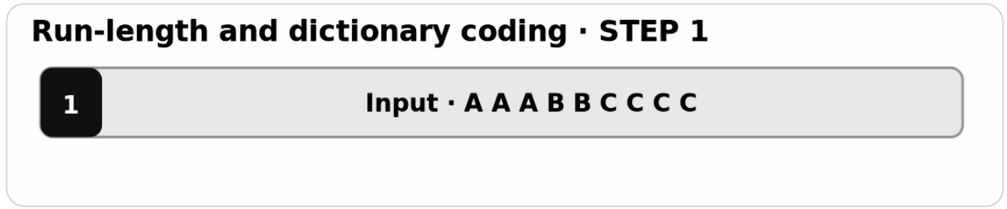
Overhead matters. Alternating characters can make naive RLE larger; long repeated phrases can be valuable for dictionary coding.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

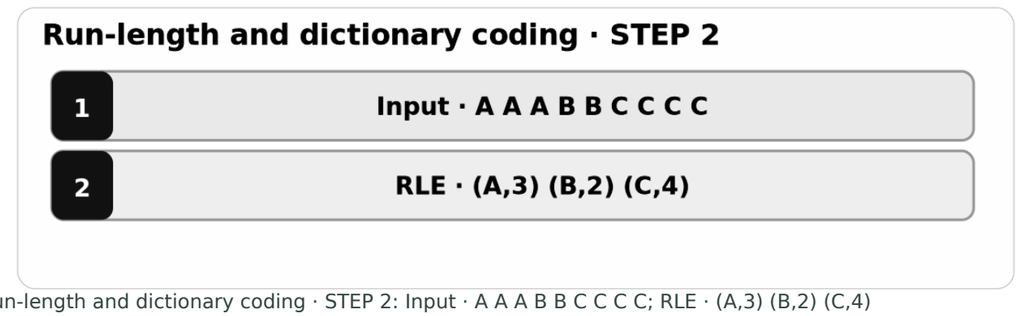
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Run-length and dictionary coding · STEP 1: Input · A A A B B C C C C

New Learning



Run-length and dictionary coding · STEP 2: Input · A A A B B C C C C; RLE · (A,3) (B,2) (C,4)

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

RLE always shortens any input file.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

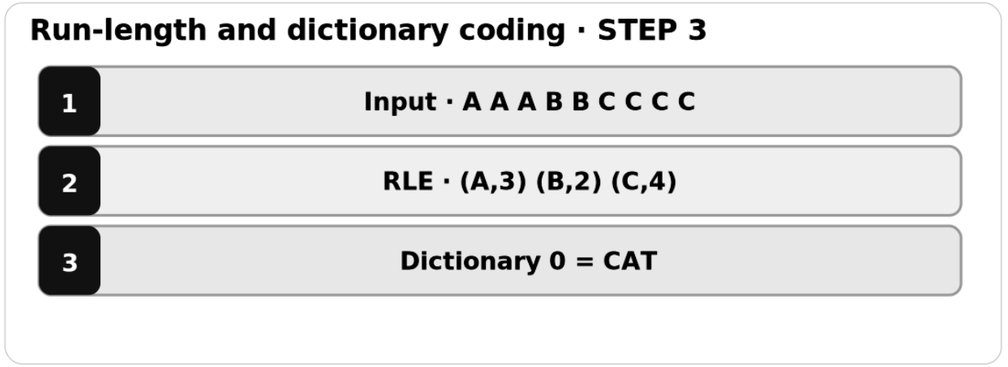
Question 3

Explain why RLE can increase size for alternating data and why dictionary coding may help repeated long substrings. [6]

New Learning · original lesson diagrams

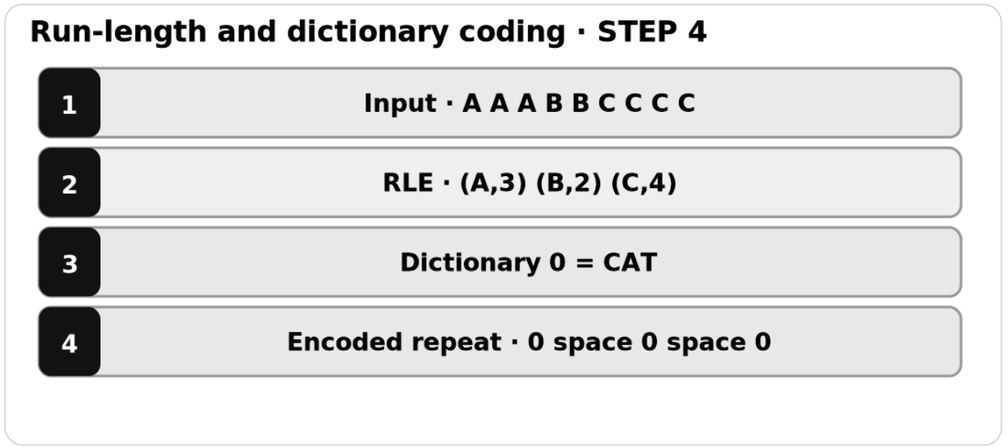
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Run-length and dictionary coding · STEP 3: Input · A A A B B C C C C; RLE · (A,3) (B,2) (C,4); Dictionary 0 = CAT

New Learning



Run-length and dictionary coding · STEP 4: Input · A A A B B C C C C; RLE · (A,3) (B,2) (C,4); Dictionary 0 = CAT; Encoded repeat · 0 space 0 space 0

I DO — WORKED EXAMPLE

Follow the worked method

Encode BBBBAAC as symbol/count pairs and verify decoding.

Predict first

Identify maximal consecutive runs: four B, two A and one C.

Model and check

Output each run as value/count and expand them back to verify exact reconstruction.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

RLE-encode AAAAABBCCCDDDD with counts. [4]

Question 2

With dictionary 0='HELLO',1='WORLD', encode 'HELLO WORLD HELLO' using references and separators. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Use dictionary reference 1 for each repeated word while preserving the space separator.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Encode BBBBAAC as symbol/count pairs and verify decoding.

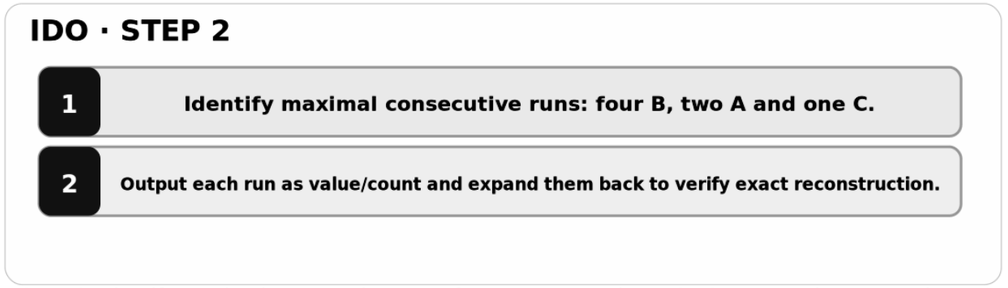
Identify maximal consecutive runs: four B, two A and one C.



IDO · STEP 1: Identify maximal consecutive runs: four B, two A and one C.

Encode BBBBAAC as symbol/count pairs and verify decoding.

Identify maximal consecutive runs: four B, two A and one C.



IDO · STEP 2: Identify maximal consecutive runs: four B, two A and one C.; Output each run as value/count and expand them back to verify exact reconstruction.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A dictionary maps index 1 to 'RED'. Encode 'RED RED' and decode the result.

Use dictionary reference 1 for each repeated word while preserving the space separator.



WEDO · STEP 1: Use dictionary reference 1 for each repeated word while preserving the space separator.